N5

NATIONAL

QUALIFICATIONS

Mark

N5/A2

Mathematics

Paper 2

(Calculator)

1 HOUR 30 MINUTES

Prelim Practice A

Fill in these boxes and read what is printed below

Forename(s)

Surname

Teacher

Total Marks - 50

Attempt ALL questions.

You may use a calculator.

To earn full marks you must show your working in your answers.

State the units for your answer where appropriate.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting.

Use **blue** or **black** ink.

Before leaving the examination room you must give this book to the Invigilator; if you do not, you may lose all the marks for this paper.

FORMULAE LIST

The roots of $ax^2 + bx + c = 0$ are $x = \dfrac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$

Sine rule: $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$

Cosine rule: $a^2 = b^2 + c^2 - 2bc\cos A$ or $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$

Area of a triangle: $A = \frac{1}{2}ab\sin C$

Volume of a sphere: $V = \dfrac{4}{3}\pi r^3$

Volume of a cone: $V = \dfrac{1}{3}\pi r^2 h$

Volume of a pyramid: $V = \dfrac{1}{3}Ah$

Standard deviation: $s = \sqrt{\dfrac{\sum(x - \bar{x})^2}{n - 1}}$

or $s = \sqrt{\dfrac{\sum x^2 - \dfrac{(\sum x)^2}{n}}{n - 1}}$, where n is the sample size.

MARKS

Total marks — 50
Attempt ALL questions

1. The value of an antique watch appreciates in value by 2.4% each year.

In 2024 it was worth £3,700.

How much will it be worth in 2028? **3**

2. $E = mc^2$

Find the value of E when $m = 3.6 \times 10^{-2}$ and $c = 3 \times 10^8$.

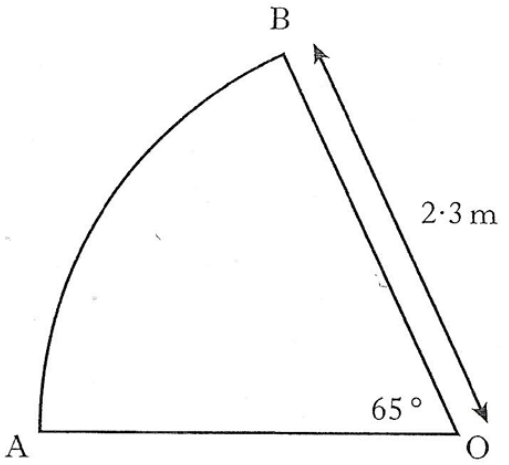
Give your answer in **scientific notation.** **3**

3. Expand fully and simplify $x(x-1)^2$. **2**

4. A sector of a circle, centre O, is shown below.

The radius of the circle is 2.3 metres.

Angle AOB is 65°.



Find the length of arc AB. **3**

5. Change the subject of the formula $p = q + 2r^3$ to r. **3**

6. Simplify $\frac{x^2-49}{x^2-5x-14}$ **3**

7. The marks of a group of students in their October tests are as follows:

41 56 68 59 43 37 70 58 61 47 75 66

a) Calculate the median and the interquartile range. **3**

The teacher arranges extra homework classes for the students before the next test in December.

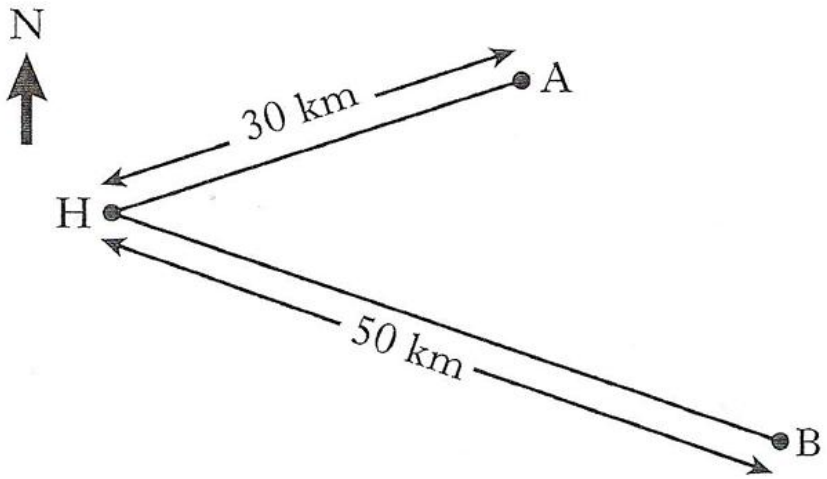
In this test, the median is 67 and the interquartile range is 14.

a) Make **two** comments comparing the marks in the October and December tests. **2**

8. Two yachts leave from harbour H to search for a whale after a reported sighting.

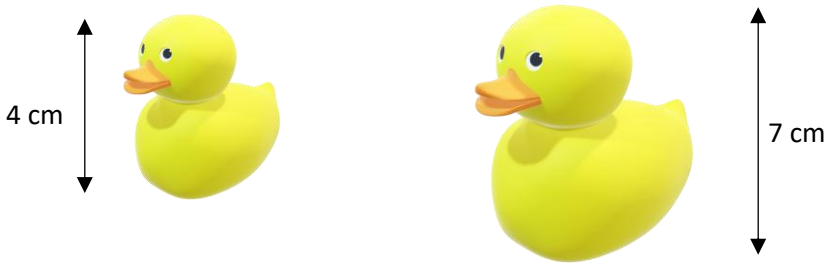
Yacht A sails on a bearing of 072° for 30 kilometres and stops.

Yacht B sails on a bearing of 140° for 50 kilometres and stops.



Calculate how far apart the two yachts are when they stop **4**

9. The two rubber ducks below are solid and mathematically similar.



The smaller duck has a height of 4 centimetres, and the larger duck has a height of 7 centimetres.

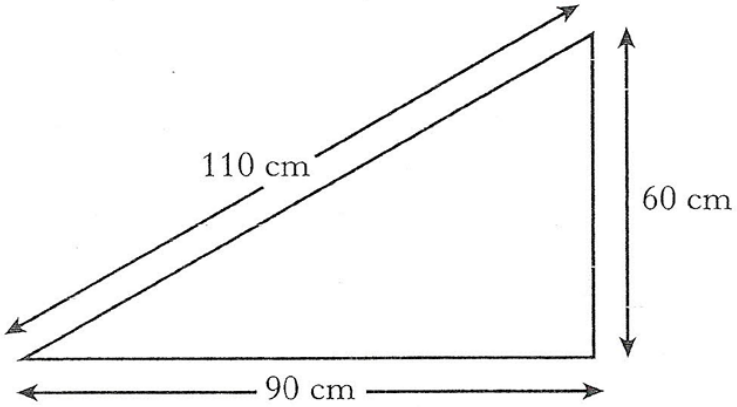
The smaller duck has a volume of 50 cubic centimetres.

Calculate the volume of the larger duck. **3**

10. After a 7% pay increase, the salary of an office worker is £38,520 per year.

Calculate the salary of the office worker before the pay increase. **3**

11. A triangular paving slab has measurements as shown.

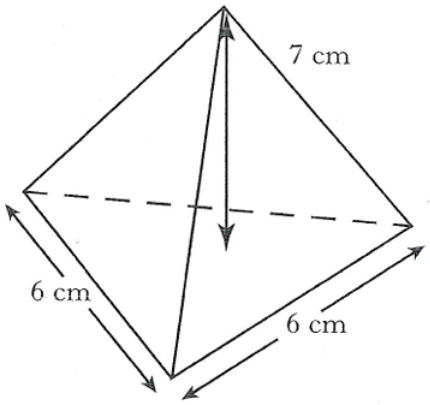


Is the slab in the shape of a right angle?

Show your working. **3**

12. Solve $\frac{4-3x}{2} - 1 > 7$ **3**

13. The diagram below shows a pyramid.

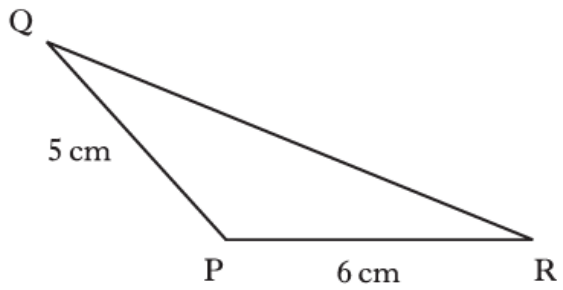


The base of the pyramid is an **equilateral** triangle of side length 6 centimetres.

The height of the pyramid is 7 centimetres.

Calculate the volume of the pyramid. Give your answer correct to **three significant figures**. **5**

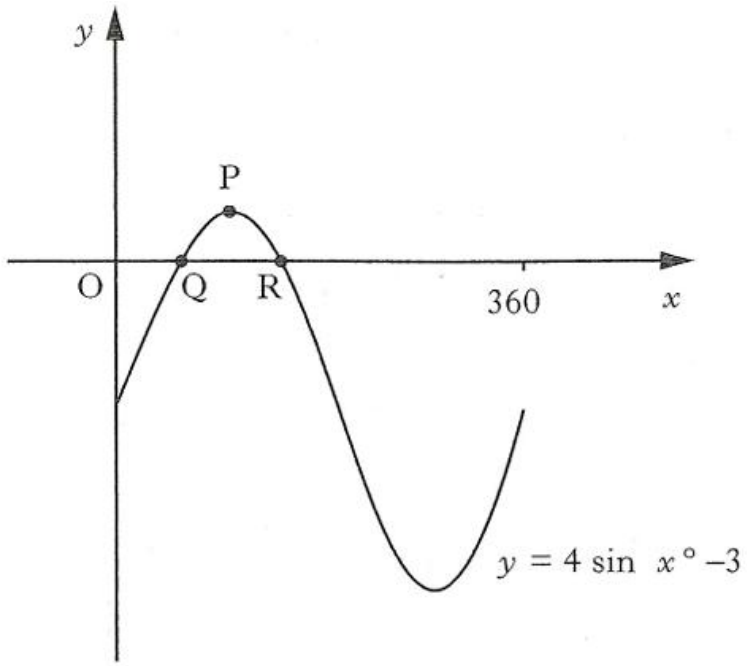
14. In triangle PQR:



- PQ = 5 centimetres
- PR = 6 centimetres
- The perimeter of triangle PQR is 21 centimetres.

Calculate the size of angle QPR. **4**

15. Part of the graph of $y = 4\sin x° - 3$ is shown below.



The graph cuts the x-axis at the points Q and R.

P is the maximum turning point.

a) Write down the coordinates of P. **2**

Point Q has coordinates $(49°, 0)$, when measured to the nearest degree.

b) State the coordinates of point R. **1**

[END OF QUESTION PAPER]